

APM348: SIR Disease Model Using Markov Chains

A Model of Infectious Disease Dynamics in Hubs

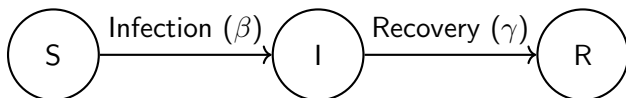
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Classical SIR Model Recap

- ▶ The standard SIR model partitions a closed population into:
 - ▶ Susceptible (S)
 - ▶ Infected (I)
 - ▶ Recovered (R)
- ▶ Dynamics are given by:

$$S' = -\frac{\beta}{N}IS, \quad I' = \frac{\beta}{N}IS - \gamma I, \quad R' = \gamma I.$$

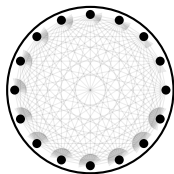


Uniform mixing assumed.

Why a New Model?

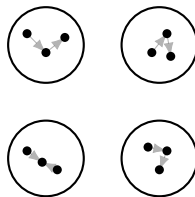
- ▶ In reality, interactions are localized (e.g., schools, homes, gyms).
- ▶ Our model introduces **hubs** to capture these localized interactions.
- ▶ Individuals interact within hubs and then move between hubs according to fixed probabilities.

Uniform Mixing



Everyone interacts with everyone.

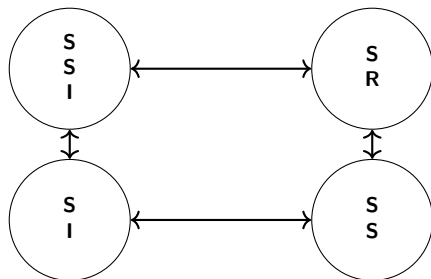
Localized Interactions (Hubs)



Individuals mostly interact within hubs.

Overview of the Hub-Based Model

- ▶ **Population Partitioning:** Individuals are distributed among a finite set of hubs.
- ▶ **Local Dynamics:** Within each hub, the disease spreads following standard SIR rules.
- ▶ **Time Structure:** Each time unit (e.g., one hour) consists of continuous interactions followed by instantaneous movement.
- ▶ **Individual Movement:** Each individual i follows a transition matrix $P^{(i)}$ determining movement probabilities.



Notation and State Description

- ▶ Let N be the set of individuals and H the set of hubs.
- ▶ For each individual i :
 - ▶ Epidemiological state: $X_i(t) \in \{S, I, R\}$.
 - ▶ Location: $L_i(t) \in H$.
 - ▶ Movement is governed by a matrix $P^{(i)}$ (with $P_{jk}^{(i)}$ for moving from hub j to k).
- ▶ For each hub h , we track:

$$S_h(t), \quad I_h(t), \quad R_h(t), \quad N_h(t) = S_h(t) + I_h(t) + R_h(t).$$

- ▶ The overall state at time t is:

$$\Phi(t) = \{(X_i(t), L_i(t)) : i \in N\}.$$

Dynamics within Each Hub

- ▶ **Infection:** In hub h , the instantaneous risk is:

$$\lambda_h(t) = \beta_h \frac{I_h(t)}{N_h(t)}.$$

- ▶ **Infection Probability:**

$$1 - \exp(-\lambda_h(t)).$$

- ▶ **Recovery:** Infected individuals recover with probability:

$$1 - \exp(-\gamma).$$

- ▶ **Assumptions:** A susceptible does not recover within a time step, and recovered individuals remain recovered.

Movement and State Transitions

- ▶ At the end of each time unit, an individual at hub j moves to hub k with probability:

$$P(L_i(t+1) = k \mid L_i(t) = j) = P_{jk}^{(i)}.$$

- ▶ This movement is memoryless—each move depends solely on the current hub.
- ▶ The overall state transition is given by:

$$P(\Phi(t+1) = (x', \ell') \mid \Phi(t) = (x, \ell)) = \prod_{i \in N} P_{\ell_i \ell'_i} M_{\ell_i}(t),$$

where $M_{\ell_i}(t)$ encodes the local health dynamics.

Local Health Dynamics via $M_h(t)$

- ▶ For each hub h , health transitions follow a matrix:

$$M_h(t) = \begin{pmatrix} e^{-\lambda_h(t)} & 1 - e^{-\lambda_h(t)} & 0 \\ 0 & e^{-\gamma} & 1 - e^{-\gamma} \\ 0 & 0 & 1 \end{pmatrix}$$

- ▶ Rows: current state $x_i(t) \in \{S, I, R\}$ Columns: next state $x_i(t+1) \in \{S, I, R\}$
- ▶ Interprets stochastic transitions:
 - ▶ $S \rightarrow I$ with infection probability $1 - e^{-\lambda_h(t)}$
 - ▶ $I \rightarrow R$ with recovery probability $1 - e^{-\gamma}$
 - ▶ $R \rightarrow R$ with probability 1

Objectives of the Model

- ▶ **Long-Term Outcome:** Determine the eventual fractions of S , I , and R , i.e.,

$$E\left(\lim_{t \rightarrow \infty} R(t)\right).$$

- ▶ **Hub Effects:** Identify hubs that act as super-spreaders.
- ▶ **Movement Patterns:** Examine how the structure of $P^{(i)}$ affects outcomes.
- ▶ **Simplification:** Assess whether a single-population SIR model can approximate the hub-based dynamics.

Simulation

Markov Chain: Definition

- ▶ A **Markov chain** is a sequence of random states $(X_0, X_1, X_2, \dots)$.
- ▶ **Memoryless Property:** The next state depends only on the current state, not on the history.
- ▶ Formally, for states i, j :

$$P(X_{t+1} = j \mid X_t = i, X_{t-1}, \dots, X_0) = P(X_{t+1} = j \mid X_t = i).$$

Transient and Recurrent States

- ▶ A state i is **recurrent** if, starting there, the chain returns to i with probability 1.
- ▶ If there is a nonzero chance of never returning to i , then i is **transient**.
- ▶ In our model, transient represent short term behaviour while recurrent states represent long term behaviour.

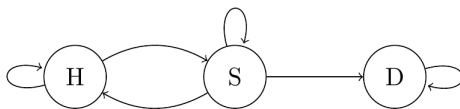


Figure: An example Markov chain

Communication Classes

- ▶ Two states i and j **communicate** if each is reachable from the other (there exist paths from i to j and from j to i).
- ▶ This communication relation is an equivalence relation, so it partitions the state space into **communication classes**.
- ▶ Each class is either entirely transient or entirely recurrent, helping us analyze long-term system behavior.

Our Model as a Markov Chain

- ▶ The overall state is $\Phi(t) = \{(X_i(t), L_i(t)) : i \in N\}$, where:
 - ▶ $X_i(t)$ is the health status (S, I, R) of individual i .
 - ▶ $L_i(t)$ is the hub location of individual i .
- ▶ Transitions occur via:
 - ▶ Local SIR dynamics within hubs.
 - ▶ Movement between hubs governed by the transition matrices $P^{(i)}$.
- ▶ The process $\{\Phi(t)\}$ is memoryless—making it a Markov chain.

Transient vs. Recurrent States in Our Model

- ▶ A state is **transient** if at least one individual is infected.
- ▶ Once the infection dies out, the system reaches a **disease-free** state, which is recurrent.
- ▶ Thus, our model eventually settles into a stable (recurrent) state with only S and R individuals.

Communication Classes in Our Model

- ▶ Two states (x, ℓ) and $(\tilde{x}, \tilde{\ell})$ communicate if:
 - ▶ They have the same health vector $x = \tilde{x}$, and
 - ▶ Their location vectors ℓ and $\tilde{\ell}$ are reachable from one another via the movement dynamics.
- ▶ Communication classes group states with identical long-term health outcomes.
- ▶ Each recurrent (disease-free) class is characterized by the final number of recovered individuals.

Goal (Markov Chain Perspective)

Goal: Determine the long-run behavior of the disease, framed in terms of the communication classes of a Markov chain Φ .

Restated Goal:

For a given integer $R(\infty) \in \{0, 1, \dots, N\}$, what is the probability that the chain Φ enters a recurrent communication class (x, ℓ) such that the health vector x contains exactly $R(\infty)$ recovered entries?

Computing the Probability Distribution (Part 1)

Step 1: Ordering the States

- ▶ List all transient states first.
- ▶ Treat recurrent communication classes as a single state.
- ▶ Follow with recurrent (disease-free) states, labeled by recovered count.

Step 2: Forming the Transition Matrix

$$A = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}.$$

- ▶ Q represents transitions among transient states.
- ▶ R represents transitions from transient to recurrent states.

Computing the Probability Distribution (Part 2)

Step 3: Fundamental Matrix

- ▶ Compute $N = (I - Q)^{-1}$.
- ▶ N gives the expected number of visits to transient states.

Step 4: Final Matrix

$$B = N R.$$

- ▶ $B_{i,r}$ is the probability of eventually reaching recurrent class r from transient state i .

Computing the Probability Distribution (Part 3)

Interpreting the Results

- ▶ Each row of B is a probability distribution over the recurrent classes.
- ▶ If row i has $B_{i,r} = 0.3$, there is a 30% chance of ending with r recovered individuals from that start.

Conclusion: By knowing the initial lumped state, the corresponding row of B gives the long-term outcome distribution.

Motivation: Excess Detail in $\Phi(t)$

- The full state

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- ▶ Once the epidemic ends, **locations no longer affect outcomes**.
 - ▶ Final outcomes depend only on the **number of recovered** individuals.
 - ▶ Many distinct $\Phi(t)$ states collapse to the **same long-term behavior**.

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 - ▶ Final outcomes depend only on the **number of recovered** individuals.
 - ▶ Many distinct $\Phi(t)$ states collapse to the **same long-term behavior**.
- ▶ Therefore, we can **group states** with the same S/I/R counts — and ignore unnecessary detail.

This motivates our definition of an equivalence relation and the quotient chain $\Psi(t)$.

A Simple Example

Consider a small system with 2 individuals and 2 hubs:

- ▶ State $\varphi_1 = ((S, 0), (R, 1))$
- ▶ State $\varphi_2 = ((R, 2), (S, 2))$

Both states have one susceptible and one recovered individual (a recovered fraction of 0.5), despite different location assignments. Thus, for our long-term analysis, these states are effectively equivalent.

Defining Equivalence of States

We define an equivalence relation $\sim$ on the state space $X = (S \times H)^N$:

$\varphi \sim \tilde{\varphi}$ if either:

1. Both φ and $\tilde{\varphi}$ are disease-free and satisfy $R(\varphi) = R(\tilde{\varphi})$; or
2. For every hub h , the triplets (S_h, I_h, N_h) are identical in both states.

This grouping means that states yielding the same overall health counts (and hence the same long-term outcome) are considered equivalent.

The Quotient Chain Ψ

- ▶ The equivalence relation $\sim$ satisfies the **strong lumping property**, ensuring that the dynamics between equivalence classes are well-defined.
- ▶ We define the quotient chain:

$$\Psi = \Phi / \sim,$$

where each state of Ψ is an equivalence class $[\varphi]$ of the original states.

- ▶ Its transition kernel is given by:

$$P\left(\Psi(t+1) = [\tilde{\varphi}] \mid \Psi(t) = [\varphi]\right) = \sum_{\psi \in [\tilde{\varphi}]} P\left(\Phi(t+1) = \psi \mid \Phi(t) = \varphi\right).$$

- ▶ This reduced chain captures the essential long-term dynamics, as the final outcome depends solely on the aggregated health counts.

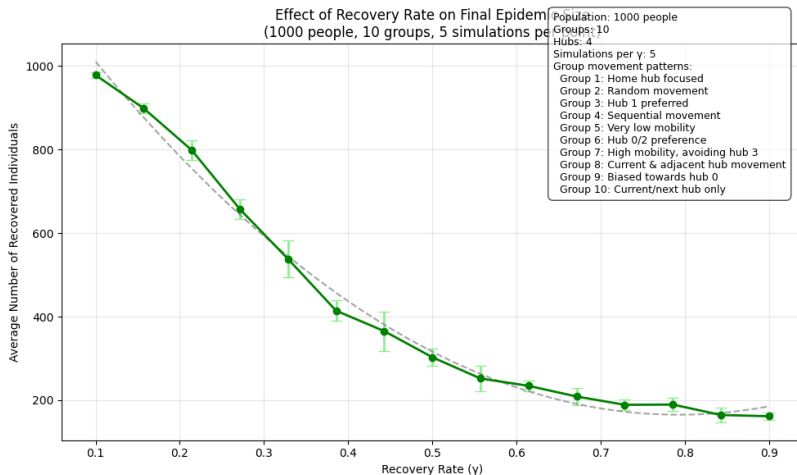
Main Theorem: Linking Ψ and Φ

Theorem: For each $R(\infty) \in [0, N]$, the probability that the quotient chain $\Psi = \Phi / \sim$ ends in the class with $R(\infty)$ recovered individuals equals the probability that Φ reaches any disease-free (recurrent) state with $R(\infty)$ recovered.

- ▶ Ψ aggregates states with the same (S, I, R) counts.
- ▶ This reduction retains the long-term outcome of Φ .

Simulation Results

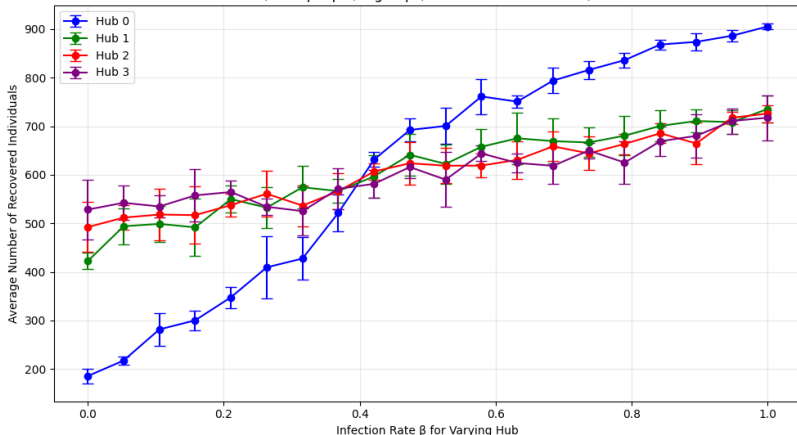
Effect of Recovery Rate γ on Epidemic Size



- ▶ Higher γ (faster recovery) leads to fewer infections.
- ▶ Consistent effect across all group-level movement strategies.

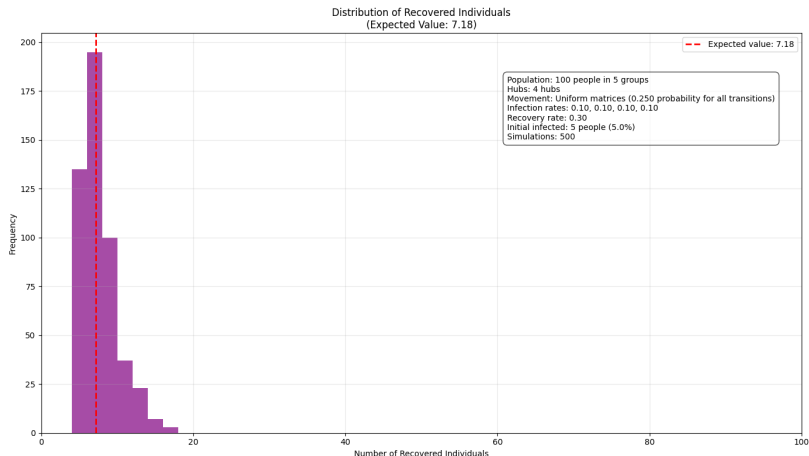
Effect of Varying β Across Hubs

Effect of Varying β at Each Hub on Final Epidemic Size
(1000 people, 5 groups, Central Hub Movement)



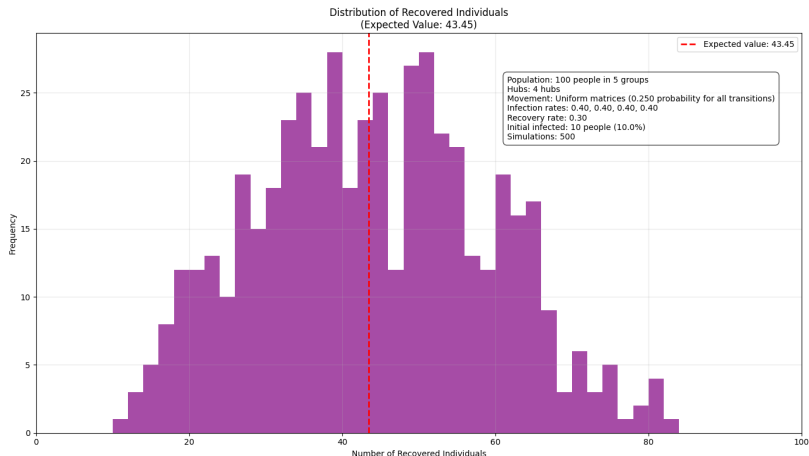
- ▶ Changing β at one hub affects outbreak size across the entire system.
- ▶ Hub 0 acts as a superspreader when its β increases.

Stochastic Behavior in Finite Populations (Low R_0)



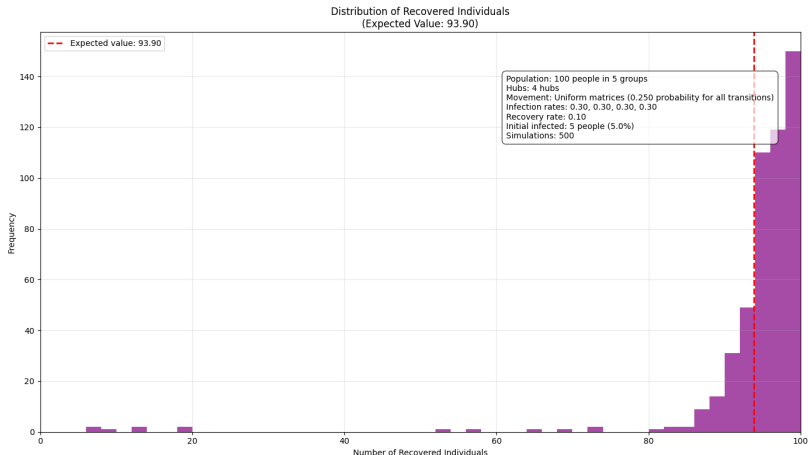
- ▶ Parameters: $\beta = 0.1$, $\gamma = 0.3 \Rightarrow R_0 = \beta/\gamma = 0.33$.
- ▶ Epidemic dies out quickly in all simulations.
- ▶ Very little variation — almost everyone remains susceptible.
- ▶ Matches the deterministic threshold behavior where $R_0 < 1$ leads to no outbreak.

Stochastic Behavior in Finite Populations (Mid R_0)



- ▶ Parameters: $\beta = 0.3$, $\gamma = 0.3 \Rightarrow R_0 = 1.0$.
- ▶ Some simulations see large outbreaks; others see very little spread.
- ▶ High stochasticity — the outcome depends sensitively on early infections.

Stochastic Behavior in Finite Populations (High R_0)



- ▶ Parameters: $\beta = 0.3$, $\gamma = 0.1 \Rightarrow R_0 = 3.0$.
- ▶ Large-scale epidemic in nearly every run.
- ▶ Low variance — most simulations lead to widespread infection and recovery.
- ▶ Final recovered count clusters around a high mean, consistent

Potential Future Analysis

- ▶ Assume that each $P^{(i)}$ is symmetric
- ▶ Assume people move along paths in a fixed deterministic cycle. Each $P^{(i)}$ would then become a permutation matrix.
- ▶ Infinite population models.
- ▶ Find more useful lumping strategies.

Conclusions

- ▶ The hub-based extension of the classical SIR model provides a more realistic description of disease spread in localized settings.
- ▶ By integrating individual movement and local infection dynamics, the model captures nuanced behavior.
- ▶ Grouping states via the quotient chain simplifies the analysis, allowing us to focus on the essential long-term outcomes.

Questions?

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